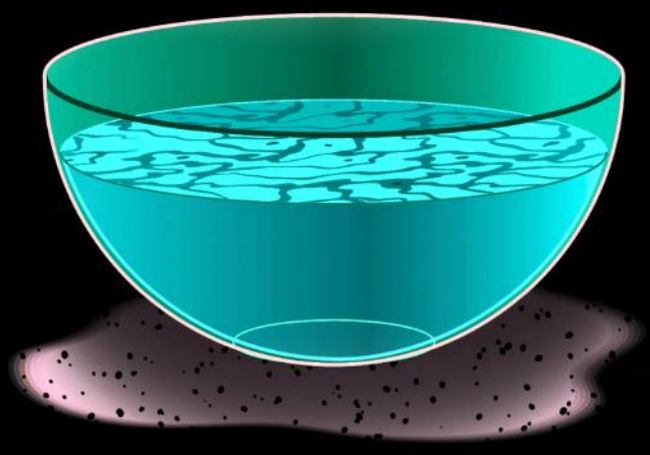
Fluid Statics:

Branch of mechanics which deals with liquids that are at rest with respect to their containers is called fluid statics (or Hydrostatics). The containers may or may not be inertial. In case of non inertial containers, generally we do analysis in container frames after applying appropriate pseudo forces.



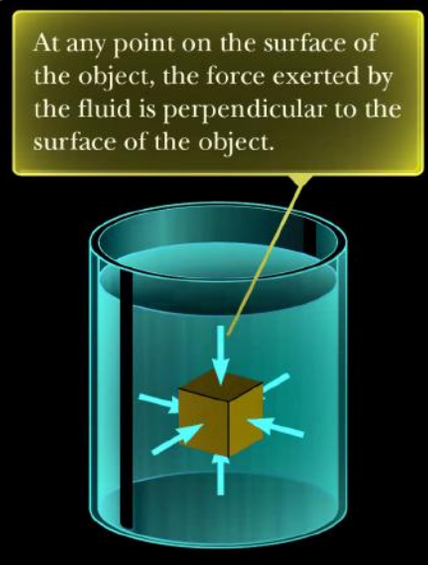
→ In Hydrostatic situations, there are no viscous forces even if the fluids themselves are sticky. This is because there are no velocity gradients and viscous forces come in to play only if there are velocity gradients (more on that in chapter on viscosity).

→ Fluids includes liquids and gases. In this chapter we will generally ignore the compressibility of fluids and assume that they have constant density (nearly) irrespective of pressure (unless specified otherwise).

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Pressure:

When we say that in hydrostatics, fluid is at rest wrt container, we mean it in a macroscopic sense. At microscopic level, fluid particles are always in random motion and will collide with any surface that comes in their way. Since on an average centre of mass of any macroscopic fluid volume is at rest, the collision forces on any such volume is exactly balanced by other forces (i.e gravity and pseudo forces). In hydrostatic situations, the force of these collisions acts perpendicular to any surface submerged in the fluid. This perpendicular force per unit area is called pressure. If we imagine an area dA submerged in a fluid on which a force dF acts then the pressure experienced



The forces exerted by a fluid on the surfaces of a sub-merged object.

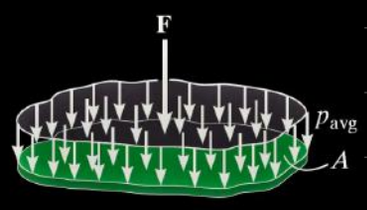
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by the lamina is $P = dF/dA$. It might



(a)

appear from definition that Pressure is a vector, however that is not true. The force experienced by surface of a submerged lamina due to collisions is independent of its orientation due to truly random motion of fluid particles. Thus pressure is independent of the orientation of the lamina and we can talk about pressure at any point in the fluid as:

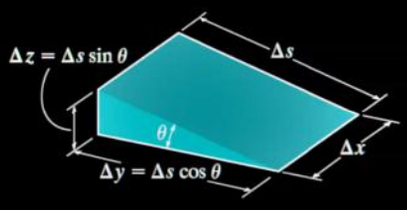


Average pressure

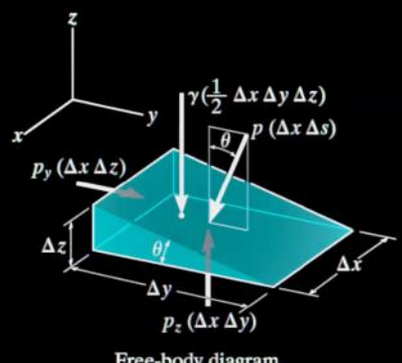
(b)

$P = |d\vec{F}/dA|$. We can also define average pressure on any lamina as the ratio of total force to the total area of the lamina. Thus $P_{avg} = F/A$. (see figure b).

PASCAL'S LAW: As stated in the last section, the intensity of the pressure acting at a point in a fluid is the same in all directions. This statement is called Pascal's law (after french mathematician Blaise Pascal : 17th century). This can be rigorously proved by considering a wedge shaped submerged element. Let P_y be the average pressure on its vertical wall, P_z be the pressure at its base and p be the average pressure at its inclined surface then, force balance gives us:



(a)



Free-body diagram

(b)

$$\Sigma F_y = 0; \quad p_y(\Delta x)(\Delta s \sin\theta) - [p(\Delta x \Delta s)]\sin\theta = 0$$

$$\Sigma F_z = 0; \quad p_z(\Delta x)(\Delta s \cos\theta) - [p(\Delta x \Delta s)]\cos\theta$$

$$-\rho g\left[\frac{1}{2}\Delta x(\Delta s \cos\theta)(\Delta s \sin\theta)\right] = 0$$

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Now rearranging y force balance equation: $p_y = p$
and rearranging z force equation, $p_z = p + \rho g \cos\theta D/2$

Now if the wedge were infinitesimal, we will get the same conclusion for local pressure also (instead of average pressures) i.e. $p_x = p_y = p_z = p$. QED. (The p_x part can be easily seen by symmetry for considering an alternative suitably oriented wedge).
Since all particles of the fluid are in equilibrium and in contact with adjacent particles due to action reaction principle (Newton's third law) any change in pressure at any point in the fluid will cause the same change at all other points within the fluid).

→ Pascal's law is also valid in accelerated vessels because acceleration can always be accomodated as pseudo gravity in the vessel frame.

STATIC PRESSURE VARIATION:

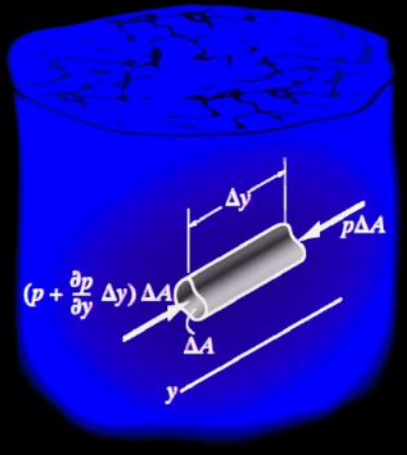
TAYLOR APPROXIMATION IN MULTIVARIABLE CALCULUS:

Just as in single variable calculus, we have the approximation $f(x+\Delta x) \approx f(x) + \frac{df}{dx}\Delta x$

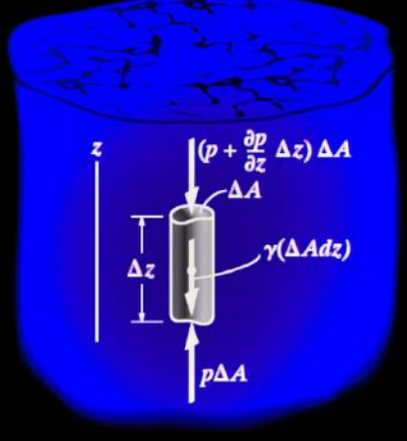
similarly for multivariable calculus we can write:

$$f(x+\Delta x, y+\Delta y, z+\Delta z) \approx f(x,y,z) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \frac{\partial f}{\partial z}\Delta z.$$

As the element becomes smaller and smaller the approximation becomes more and more accurate.



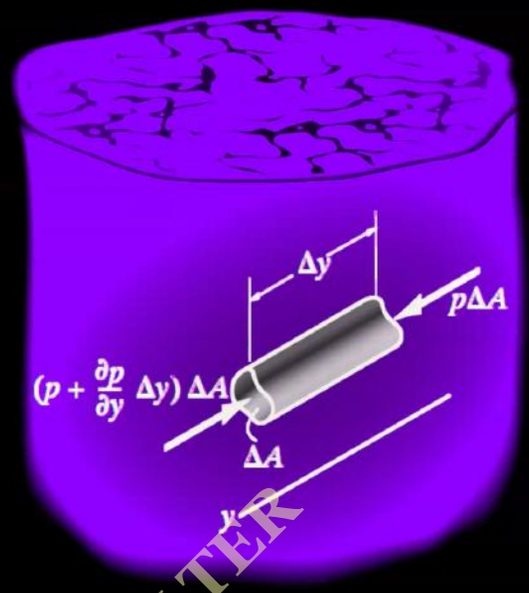
(a)



(b)

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Pressure Variation Along Horizontal:



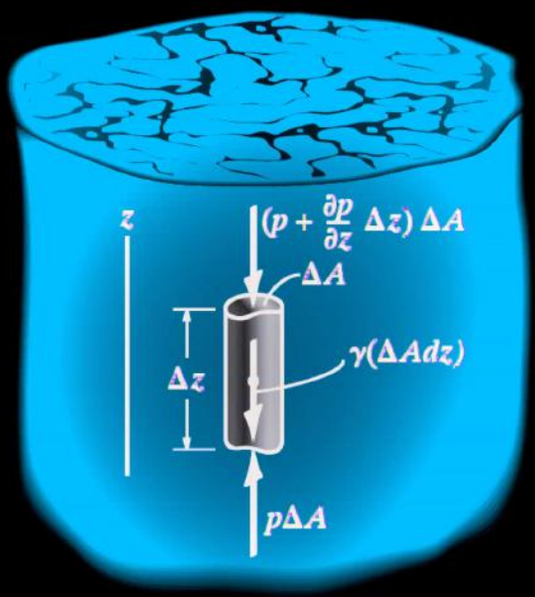
Consider a fluid element as shown in the figure. Here y direction is along horizontal. By force balance in horizontal direction:

$p\Delta A = (p+\Delta p)\Delta A$ where $\Delta p = \frac{\partial p}{\partial y}\cdot \Delta y$.

Thus $\Delta p = 0$ Thus along horizontal direction, the pressure does not change. i.e:

→ In any confined continuous body of liquid in a container, pressure at all points at same horizontal level is equal.

Pressure Variation Along Vertical:



Consider a submerged element of a fluid as shown in adjacent figure. Let γ be the specific weight of the fluid (i.e ρg). Then by vertical force balance:

$$(p+dp)\Delta A + \gamma \Delta A dz = p\Delta A \quad \left(dp = \frac{\partial p}{\partial z}\Delta z\right).$$

Thus $\frac{dp}{dz} = -\gamma$ or $dp = -\gamma dz$ or

$$\boxed{dp = -\rho g\, dz} \quad (1)$$

→ Here the negative sign signifies that pressure decreases as one moves upwards in the fluid.

→ If P_0 is atmospheric pressure the pressure in a liquid of constant density at a depth h from free surface is

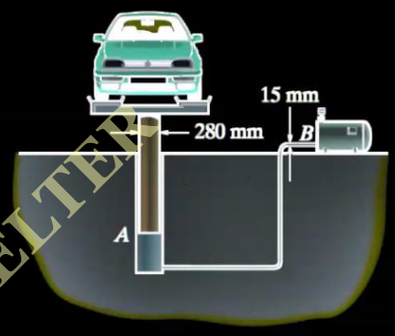
$$\boxed{p = P_0 + \rho g h}$$

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→ If density varies with depth we can write equation ① as $dp = \rho g dh$ and then integrate in appropriate limits after putting ρ as a function of depth to get the required pressure function.

EXAMPLE:

A pneumatic jack is used in a service station as shown in Figure. If the car and lift weigh 25 kN, determine the force that must be developed by the air compressor at B to raise the lift at a constant velocity. Air is in the line from B to A. The air line at B has an inner diameter of 15 mm, and the post at A has a diameter of 280 mm.



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Solution: Since specific weight of air is negligible ∴ pressure at A must be equal to pressure at B i.e $P_A = F_B/A_B$. Also, $P_A = W/A_A$. Thus:

$$\frac{W}{A_A} = \frac{F_B}{A_B} \Rightarrow F_B = W \times \frac{A_B}{A_A}$$ (Here pressure force is balancing the weight).

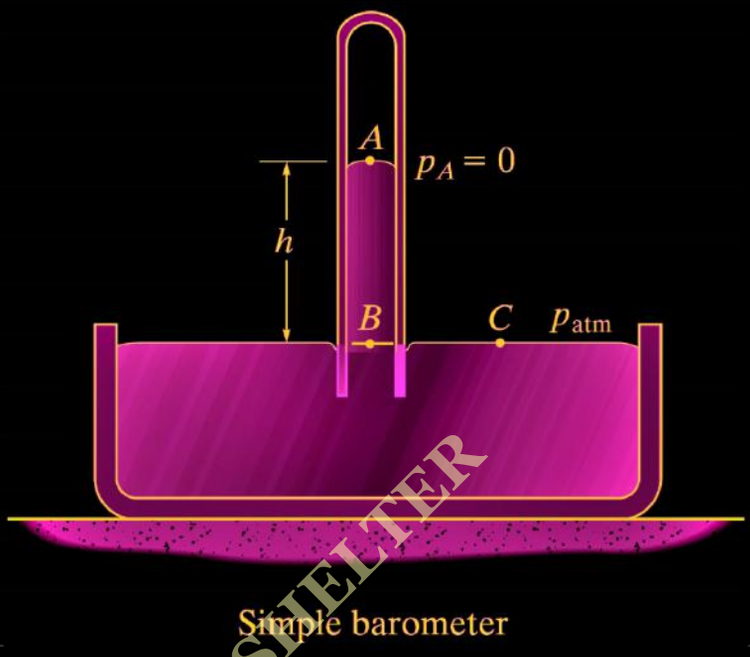
i.e $$F_B = \frac{25 \times 10^3 \times \pi (0.0075\,m)^2}{\pi \times (0.140\,m)^2} = 71.7\,N.$$

→ This might seem surprising that how a 25 kN car can be supported by applying so little force. However, this is a direct consequence of Pascal's law.

→ This does not mean any kind of violation of energy conservation. To raise the car to a certain height, piston will have to make many times more displacement (since volume of air is conserved. Thus $F_A \times h = F_B \times l$ where h is the height raised and l is cumulative displacement of piston while applying the force F_B. (∵ $A_A \times h = A_B \times l$).

Simple Barometer (Torricelli's Barometer):

A barometer is a device used to measure atmospheric pressure. The working of a simple barometer can be understood from adjacent figure. The apparatus consists of a trough containing mercury and an inverted tube which has vacuum at its upper portion and mercury in the lower portion. (This can be done by inverting a fully filled sealed tube in the trough and then removing the seal. Now let h be the height of mercury in the vertical tube above free surface of the mercury in the trough then since pressure at same level in a liquid is same, $\rho g h = p_{atm}$ where ρ is the density of mercury and p_{atm} is atmospheric pressure.



Simple barometer

→ Note that vacuum exerts zero pressure because there are no molecules to collide. Thus for a simple barometer:

$$\boxed{p_{atm} = \rho g h}$$

→ Mercury is preferred liquid for barometers due to its high density. We can rewrite the barometer equation as $h = p_{atm}/\rho g$. Thus smaller the ρ greater the h. For water the required height will be about 10 m while for mercury it is just 76 cm this is because specific gravity of mercury is 13.6 (i.e it is 13.6 times more dense than water).

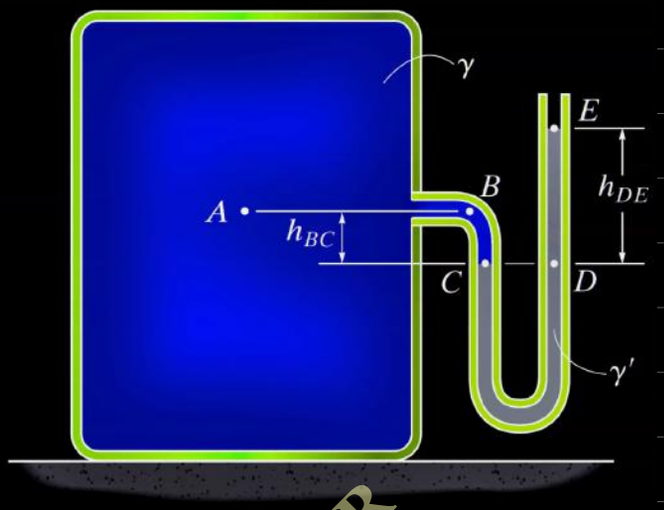
→ Sometimes we come across problems involving defective barometers in which some air is trapped at the top of the barometer. In such problems we also need to add the pressure due to the trapped air to the pressure due to weight of the mercury column. (Will be dealt in Thermodynamics)

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MANOMETER:

A manometer is used to find pressure of gases in pressurized chambers or the vacuum pressure in chambers at sub atmospheric pressures. It consists of a U shaped tube filled with liquid of certain specific weight (say γ'). Let γ be the specific weight of the gas, then equating pressure at C and D in the two limbs,

$$P_A + \gamma h_{BC} = P_{atm} + \gamma' h_{DE} \Rightarrow \boxed{P_A = P_{atm} + \gamma' h_{DE} - \gamma h_{BC}}$$



Simple manometer

Thus knowing the height of gas and liquid in the two limbs, we can find the pressure at point A. Typically in case of gases, γ is negligible and accordingly the term γh_{BC} is ignored and the pressure is simply taken as $P_{atm} + \gamma' h_{DE}$.

GAUGE PRESSURE: The pressure at any point over and above the atmospheric pressure is called gauge pressure. Thus if P is pressure at any point and P_{atm} is atmospheric pressure then, gauge pressure is simply:

$$\boxed{P_{gauge} = P - P_{atm}}$$

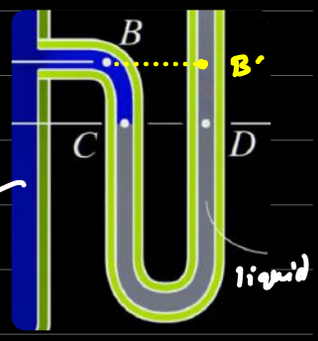
For instance in case of gases, if we use the manometer, the gauge pressure is simply $\gamma' h_{DE}$ (see figure from previous section). Thus we can read gauge pressure directly by looking at liquid level difference in the two limbs of manometer.

→ In manometer tubes, pressure at same level is same only if the same continuous liquid column exists between two points. If the liquids are different, pressures will be different at the same level (in general). For instance in manometer shown above, pressure is B is not same as pressure at same level in right

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column.

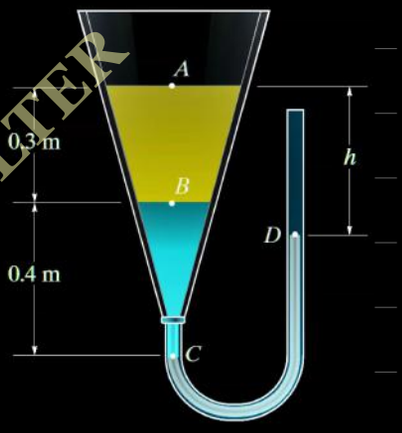
Think! Which one will have higher pressure, B or B' (say) in the right tube?



Solution: Pressure at C and D is same now since density of liquid is more than that of the gas, pressure along DB' decreases more than that of gas while going along CB. Thus B' has smaller pressure than B.

EXAMPLE:

The funnel in Figure is filled with oil and water to the levels shown, while portion CD of the tube contains mercury. Determine the distance h the mercury level is from the top of the oil surface. Take $\rho_o = 880\ \text{kg/m}^3, \rho_w = 1000\ \text{kg/m}^3, \rho_{\text{Hg}} = 13\,550\ \text{kg/m}^3$.



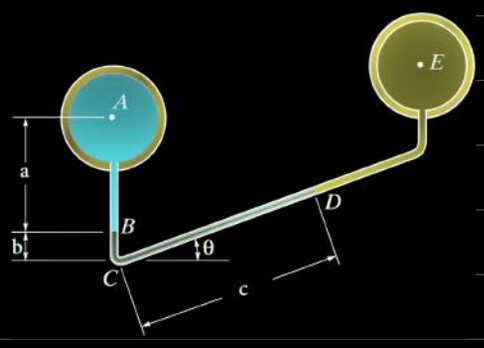
Solution: Equating the pressure at point C from left and right columns:

$$\rho_o g \times 0.3 + \rho_w \times g \times 0.4 = \rho_{Hg} g (0.3 + 0.4 - h)$$

Now cancelling g and substituting desities and solving for h, we get $h = 0.651\ m$.

EXAMPLE:

The ***inclined-tube manometer*** shown in Figure is used to measure small pressure changes. Determine the difference in pressure between points A and E if the manometer liquid, mercury, is at the level shown. The pipe at A contains water, and the one at E contains natural gas. For mercury and water densities are ρ_m and ρ_w respectively .



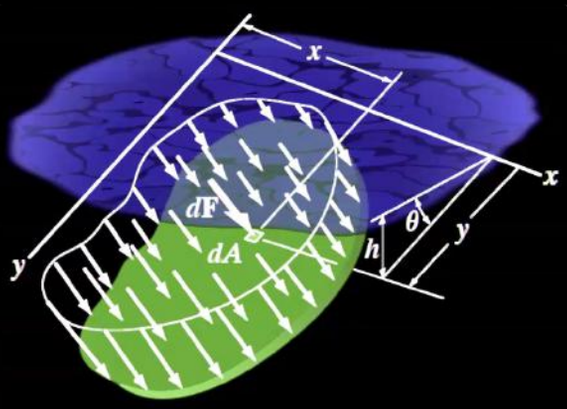
Solution: We will travel from point A to E adding all the pressure increments due to liquid columns (and ignore gas columns due to small densities). Thus:

$$P_A + \rho_w g a + \rho_m g b - \rho_m g c \sin\theta - 0 = P_E$$

$$\Rightarrow P_A - P_E = \rho_m g [c \sin\theta - b] - \rho_w g a \quad \text{Ans.}$$

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Hydrostatic Force On A Submerged Lamina:



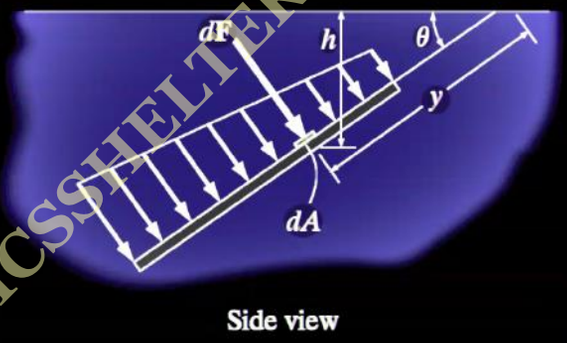
Consider a submerged lamina. We wish to find the resultant hydrostatic force due to gauge pressure on it. We draw x axis as a horizontal on water surface in plane of lamina and y axis as line of steepest slope in plane of lamina. Now force experienced by an area element dA:

$$dF_R = \rho g h \, dA = \rho g \sin\theta \, y \, dA$$

$$\Rightarrow F_R = \rho g \sin\theta \int y \, dA$$

$$= \rho g \sin\theta \cdot \left[\frac{1}{A}\int y \, dA\right] A$$

$$\Rightarrow F_R = \rho g \sin\theta \, \bar{y} \, A$$ where $\bar{y}$ is coordinate of centroid.



Side view

Thus $\boxed{F_R = P_c A}$ where P_c is gauge pressure at centroid of the area.

→ If atmospheric pressure is P_0 then net force will be given by

$\boxed{F = (P_0 + P_c) A}$ This is because at every element the net pressure is simply the gauge pressure offset upwards with atmospheric pressure.

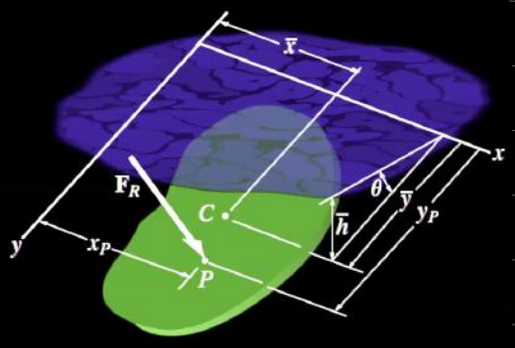
F_R: Resultant Gauge Pressure force, P_0: Atmospheric pressure.

F: Resultant Force

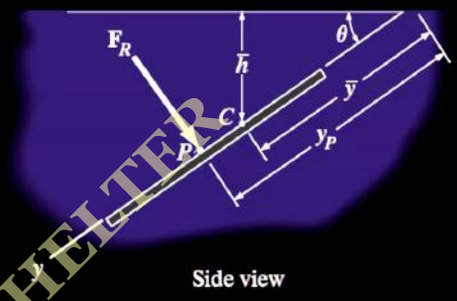
→ Centroid of an area is simply centre of mass of the area assuming its areal mass density constant.

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CENTRE OF PRESSURE:



The effective point where resultant gauge pressure can be assumed to be acting for purpose of calculating torques is called centre of pressure. For purpose of calculating torques about horizontal axes, we will be only requiring y coordinate of centre of pressure. Now by definition of centre of pressure, considering torque about x axis, if y_p is effective y coordinate of the force then:



Side view

$$y_p F_R = \int y\, dF = \int y \times \rho g y \sin\theta\, dA$$

$$\Rightarrow y_p F_R = \rho g \sin\theta \int y^2 dA$$

Now $\int y^2 dA$ is simply area moment of inertia which can be calculated simply by substituting area in the formula for mass moment of inertia. Thus:

$$y_p F_R = I_x \cdot \rho g \sin\theta \Rightarrow y_p \times \rho g \bar{y} \sin\theta \times A = I_x \rho g \sin\theta$$

$$\Rightarrow y_p = \frac{I_x}{A\bar{y}} = \frac{k^2}{\bar{y}}$$

(I_x: Area moment of inertia, k^2: Radius of gyration about surface axis, $\bar{y}$: y coordinate of centroid).

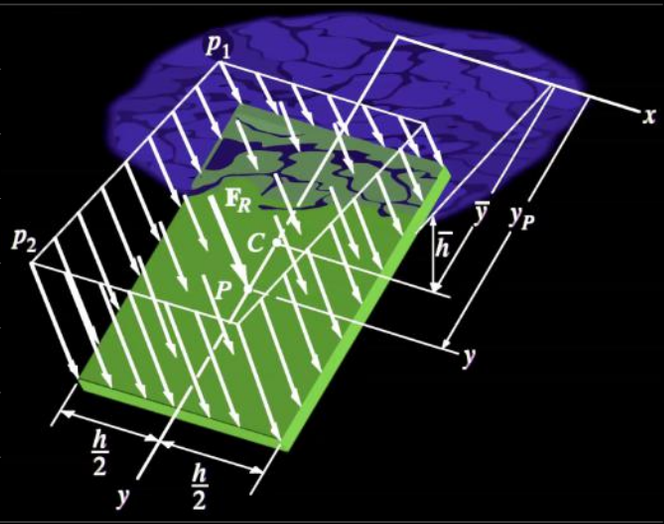
Thus $\boxed{y_p = \dfrac{I_x}{A\bar{y}} = \dfrac{k^2}{\bar{y}}}$ Also, $\underbrace{I_x = \bar{I}_x + A\bar{y}^2}_{\text{Parallel Axis Theorem.}} \Rightarrow \boxed{y_p = \dfrac{\bar{k}^2}{\bar{y}} + \bar{y}}$

→ A similar formula could also be derived for finding x_p but it involves product of inertia. So if torque is required about any non-horizontal axis (parallel to lamina), we will use direct integration of elemental torques.

→ In case lamina is symmetric about a line of steepest descent, then centre of pressure lies on the symmetry axis itself.

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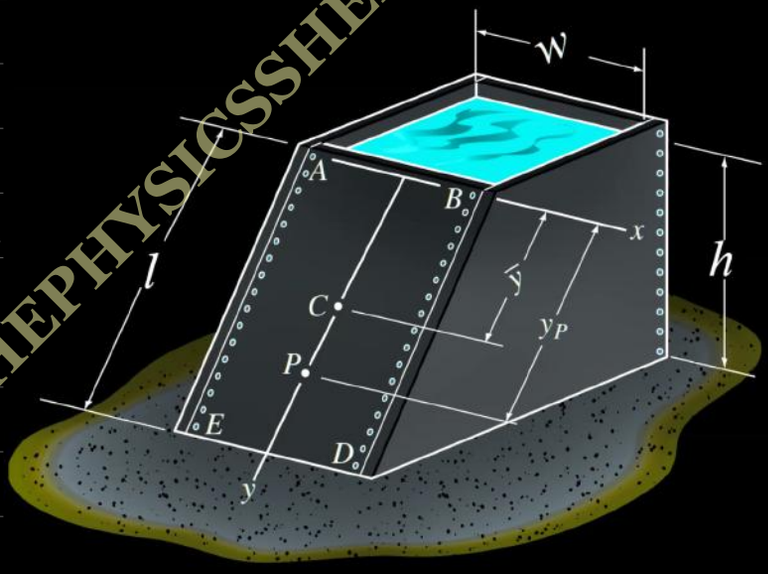
→ For instance for the shown rectangular lamina, $x_p = 0$ (i.e line of symmetry contains centre of pressure). Here we can easily calculate force and torque using $\bar{y}$ and y_p.



→ In case torque due to net pressure forces is required we put an additional $P_0 A$ at centroid of the area ($\because P_0$ is a uniform field).

EXAMPLE:

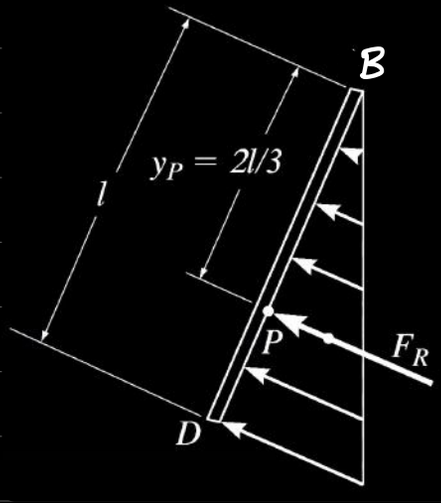
For the shown figure find (a) Net resultant gauge pressure force (b) Effective y_p of gauge pressure force (c) Torque of gauge pressure force about line ED.



Solution: $\bar{y} = \frac{l}{2}$ Thus, gauge pressure force : $F_R = \rho g \frac{h}{2} \times wl$

i.e $F_R = \rho g h w l / 2$ Ans.

Now $y_p = k^2/\bar{y} = [l^2/3]/[l/2] = 2l/3$ Ans.



Now torque of Gauge pressure force about line ED:

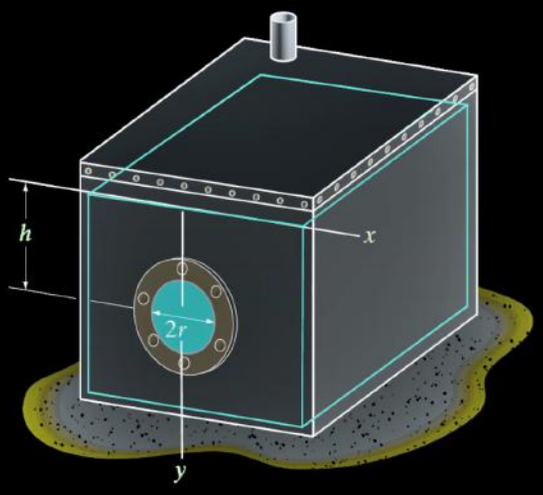
$\tau = F_R \times (l - y_p) = \frac{\rho g h w l}{2} \times (l/3) = \rho g h w l^2/6$ Ans.

→ In most situations we are only interested in gauge pressure torques because P_0 offset acts from both inside and outside the surface causing a net zero torque when both sides of the vessel walls are considered.

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EXAMPLE:

Determine the magnitude and location of the resultant force that the water pressure exerts on the circular plate shown in the figure.



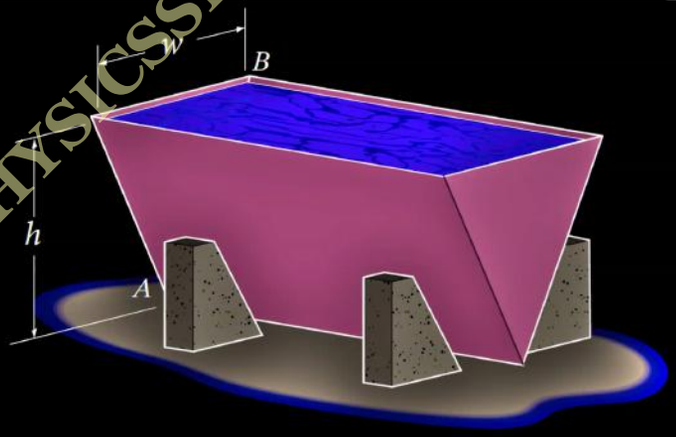
Solution: $F_R = A\rho g\bar{h} = \rho g h \times \pi r^2$

Thus $F_R = \pi \rho g h r^2$ Ans.

Now $y_p = h^2/\bar{y} = I_x/A\bar{y} = A(r^2/4 + h^2)/(Ah) = \frac{r^2}{4h} + h$ Ans.

EXAMPLE:

Determine the magnitude and location of the resultant force acting of triangular end plate of the settling tank. Fluid density is ρ.

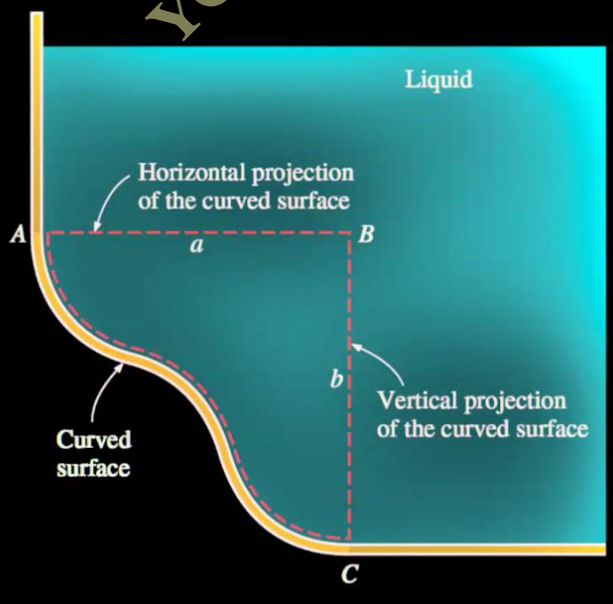


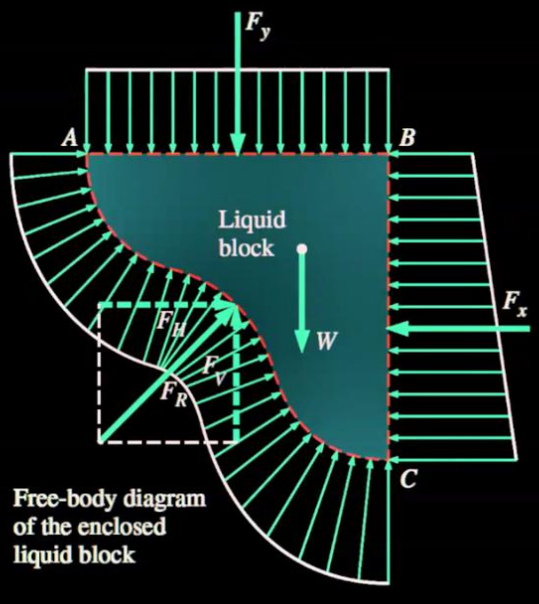
Solution: $F_R = \rho g \bar{y} A = \rho g \frac{h}{3}(wh/2)$

$\Rightarrow F_R = \rho g h^2 w/6$ Ans. Now $y_p = I_x/A\bar{y} = (Bh^3/6)/(Bh/3) = h/2$ Ans.

FORCE ON CURVED OR INCLINED SURFACES BY PROJECTIONS:

CASE I: LIQUID ABOVE THE SURFACE:





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Consider the free body diagram of the liquid volume inside the dotted boundary. Here for equilibrium, Vertical force offered by the curved surface to water must be equal to the weight of the water + F_y i.e the vertical force due to fluid above this volume thus:

$$\boxed{F_V = F_y + W}$$ ← Vertical force on curved/inclined surface. (Surface below the water volume).

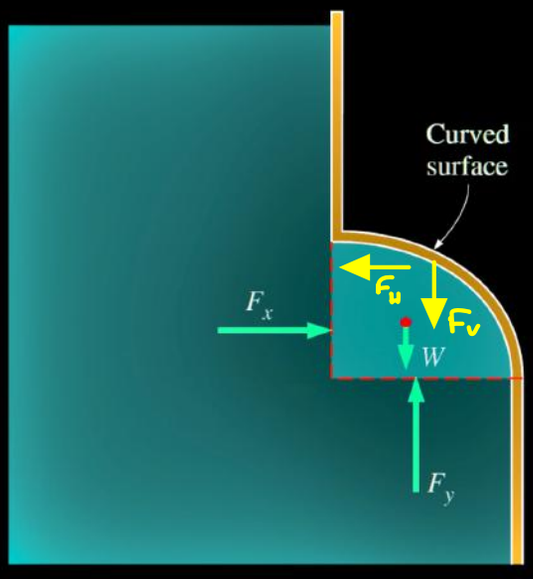
Now for horizontal equilibrium of fluid volume, Force on horizontal projection surface must exactly balance the force offered by the curved surface in horizontal direction. Thus:

$$\boxed{F_H = F_x}$$ ← Horizontal force (see figure).

CASE II: LIQUID BELOW THE LIQUID SURFACE:

In this case weight of water acts vertically away from the curved surface while force due to remaining fluid acts upwards. (see figure). Thus here, $F_v + W = F_y$

$$\Rightarrow \boxed{F_v = F_y - W}$$



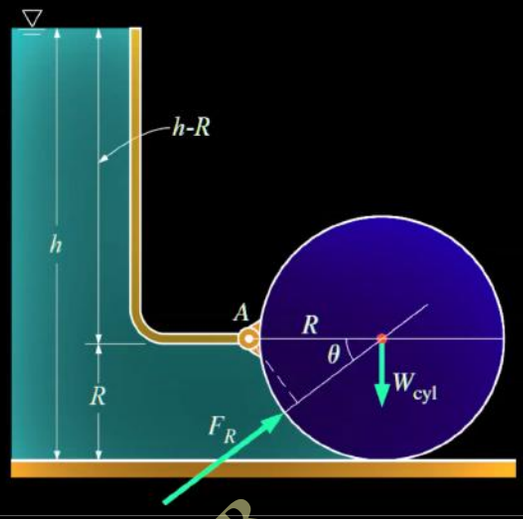
The horizontal force calculation remains similar to the previous case.

→ Thus in case of curved surfaces, we can find force by projections. Torque about any specific axis of water forces can also be found by considering rotational equilibrium of the fluid element under action of F_H, F_y weight and forces due to curved surface.

→ In case of cylindrical surfaces / spherical surfaces, torque of hydrostatic forces about axis of curvature / centre of curvature will be zero because all forces pass through it.

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EXAMPLE: The height of a water reservoir is controlled by a cylindrical gate hinged to the reservoir. When the height in the reservoir becomes h, the gate opens by turning about the hinge A. Find [(a) Hydrostatic force and its line of action when the gate opens. (b) Weight of the cylinder] per unit length.

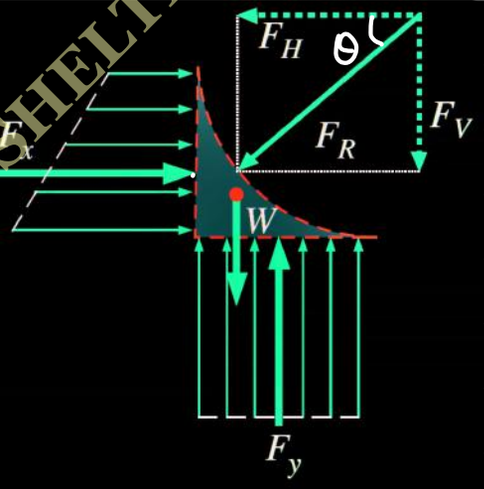


Solution: Let length of cylinder be ℓ. Vertical force on cylinder:

$$F_V = \rho g h \times R\ell - (R^2 - \pi R^2/4)\ell\rho g$$

$$F_V = \rho g R \ell \left[h - R + \pi R/4\right]$$

$$F_H = \rho g (h - \tfrac{R}{2}) R\ell$$



Now at the verge of turning, vertical hinge force must balance weight of cylinder and F_V

Thus vertical hinge force:

$$F_{Ay} = \rho g R \ell \left[h - R + \frac{\pi R}{4}\right] - W_{cyl}.$$

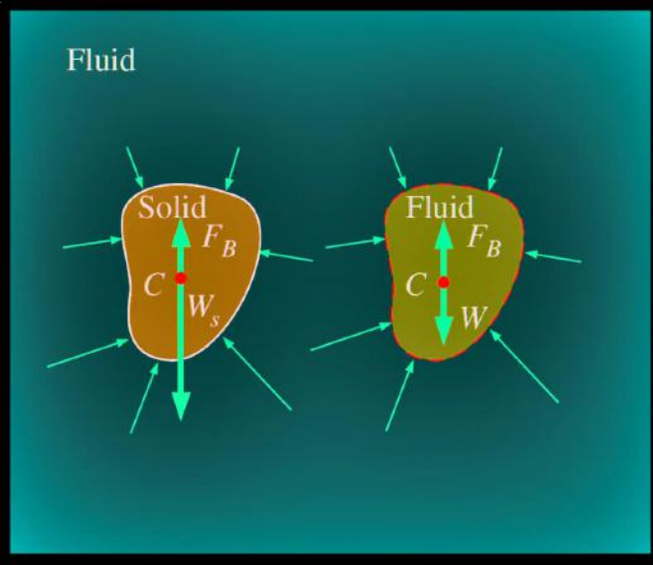
Now at the verge of turning net torque about CM of cylinder must be zero. Also Weight, hydrostatic force and horizontal hinge forces have zero torques, therefore, vertical hinge force must also have zero torque. This can happen only if the vertical hinge force is itself zero (because its arm is non-zero). Thus $F_{Ay} = 0$

$$\Rightarrow \quad W_{cyl}/\ell = \rho g R \left[h - R + \pi R/4\right] \quad \text{Ans.}$$

→ Notice that vertical force due to hinge would have been zero even before reaching a critical height of the liquid in reservoir (as clear from torque argument). At this stage, Normal reaction would balance the other vertical forces on the cylinder.

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ARCHIMEDES PRINCIPLE: The Buoyant force (Resultant of all pressure forces) on a body immersed in a fluid is equal to the weight of fluid displaced by the body and it acts upwards through the centroid of the displaced volume.



PROOF: Consider an arbitrarily shaped solid body submersed in a fluid at rest and compare it to a body of fluid of same shape indicated by dashed red line (see figure). The buoyant forces on these bodies are same since the pressure distributions which depend on depth of various points on bodies from free surface are same. Now the imaginary fluid is in static equilibrium. Thus buoyant force on it must be equal to its weight and must pass through its centre of mass (for torque balance). Thus the upward buoyant force on the solid body must also be equal to the weight of the imaginary fluid whose volume is same as that of the solid body and must have same force distribution profile and thus similar line of action (i.e passing through centre of mass of the displaced liquid).

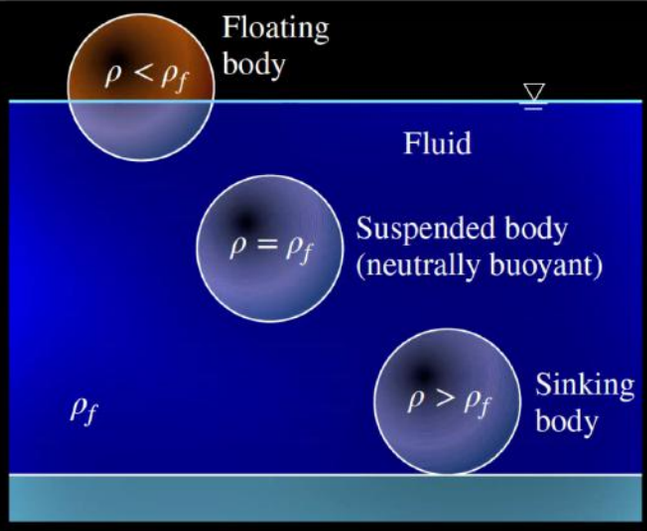
CENTRE OF BUOYANCY: The centre of mass of the displaced liquid volume is called centre of buoyancy. This is the point at which Buoyant force may be considered to be concentrated for purpose of calculating the torques.

→ Centre of Buoyancy has got nothing to do with centre of mass of the submersed solid body. Thus even if submersed solid body were non-homogeneous, we will assume a homogeneous volume of fluid in its place and then calculate centre of buoyancy.

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Floating / Sinking And Neutral Buoyancy:

Depending on relative magnitudes of average density of a solid and the density of a fluid, the body may fall in to one of the three cases as shown in the adjacent figure.

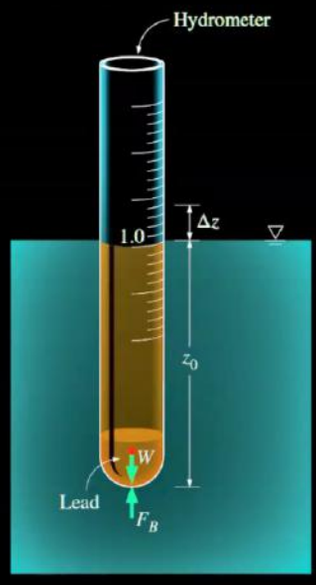


Specific Gravity:

The ratio of density of a given substance to that of water is called its specific gravity. For instance specific gravity of mercury is 13.6 which means its density is 13.6×10^3 kg/m^3.

Example:

If you have a seawater aquarium, you have probably used a small cylindrical glass tube with a lead-weight at its bottom to measure the salinity of the water by simply watching how deep the tube sinks. Such a device that floats in a vertical position and is used to measure the specific gravity of a liquid is called a *hydrometer* (See figure). The top part of the hydrometer extends above the liquid surface, and the divisions on it allow one to read the specific gravity directly. The hydrometer is calibrated such that in pure water it reads exactly 1.0 at the air–water interface. (*a*) Obtain a relation for the specific gravity of a liquid as a function of distance Δz from the mark corresponding to pure water and (*b*) determine the mass of lead that must be poured into a 1-cm-diameter, 20-cm-long hydrometer if it is to float half-way (the 10-cm mark) in pure water. (NEGLECT WEIGHT OF THE GLASS TUBE, Also neglect curvature of the bottom of the tube)



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Solution: If A is the cross section of hydrometer and ρ_w density of water, then: $\rho_w A z_0 g = mg$ ① (m: mass of hydrometer).
Similarly if ρ is density of another fluid and $z_0 + \Delta z$ total submersed depth then: $\rho A (z_0 + \Delta z) g = mg$ ② Now from ① & ②:

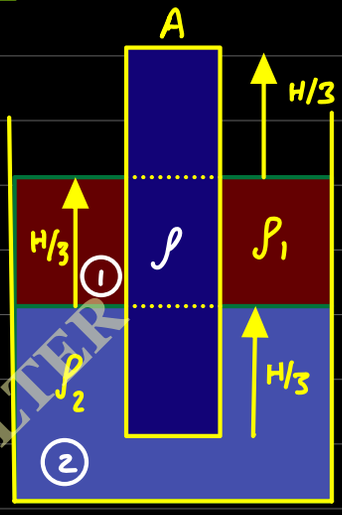
$$\rho (z_0 + \Delta z) = \rho_w z_0 \Rightarrow S = \rho/\rho_w = z_0/(z_0 + \Delta z) \text{ Ans.}$$

(b) For half submergence, $F_B = mg$ i.e $\rho_w \frac{l}{2} . Ag = mg$

(l: total height of hydrometer). Thus $m = \rho_w l A/2$

$= 1000 \times 0.2 \times \pi (0.005)^2/2 = 0.00785\ Kg = 7.85\ g$ Ans.

Buoyant Force On A Solid Crossing Multiple Liquids:



Consider a solid prism of cross section A being submerged as shown in the figure. Here by drawing the FBD of displaced fluids it can be shown that net buoyant force is equal to net weight of all displaced liquids even though individual forces by liquids are not equal to their displaced weights. For instance in this case, Net Buoyant force is $(\rho_1 A H/3) g + \rho_2 (A H/3) g$ however

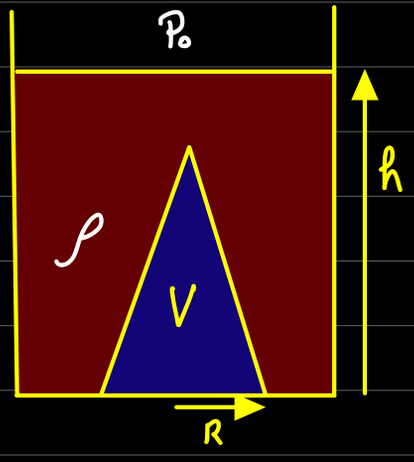
individual force due to liquid ① is zero and that due to liquid ② is $(\rho_1 g H/3 + \rho_2 g H/3) A$. (We are considering only gauge pressure force here).

Think! What is the density of the prism in previous situation.

Solution: By force balance, $\rho A H g = \rho_1 A H/3\, g + \rho_2 A H/3\, g$

$\Rightarrow \rho = (\rho_1 + \rho_2)/3$ Ans.

Submerged Bodies With Water Tight Surfaces:



While considering the fluid force on submerged bodies having water-tight surfaces, one must take in to account the fact that there will be no fluid force acting on water-tight surfaces. For instance, in adjacent figure, the bottom surface of the cone will experience no fluid pressure if water does not get under it. Thus for finding fluid force we can't just use the buoyant force formula. We must compensate for the extra counted missing force on bottom surface. Thus here correct expression for fluid force on cone will be

$$F_F = \rho V g \uparrow - (\rho g h + P_0) \times \pi R^2 \uparrow$$

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Thus $F_F = P_0 \pi R^2 + \rho g [\pi R^2 h - V]$ ↓

(Note that here net force will be downwards since entire curved surface of the cone is being pushed normally, causing a net downward component of the fluid force).

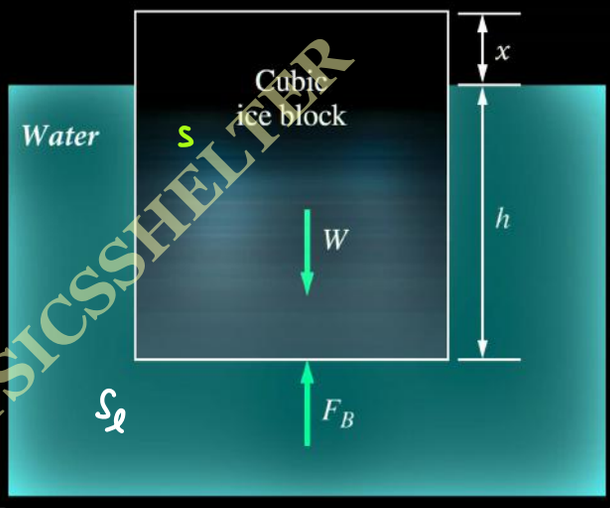
Fractional Submerged Volume In Terms Of Specific Gravity:

Let V be the total volume of a floating body having specific gravity s and let specific gravity of the liquid be S_ℓ. We wish to calculate the fraction of the total volume submersed in equilibrium. By vertical force balance:

$\rho_w S_\ell \eta V g = \rho_w S V g$ where η is the submersed fraction. Thus:

$$\boxed{\eta = S/S_\ell}$$

in case of water, $S_\ell = 1$ thus for water, $\eta = S$.

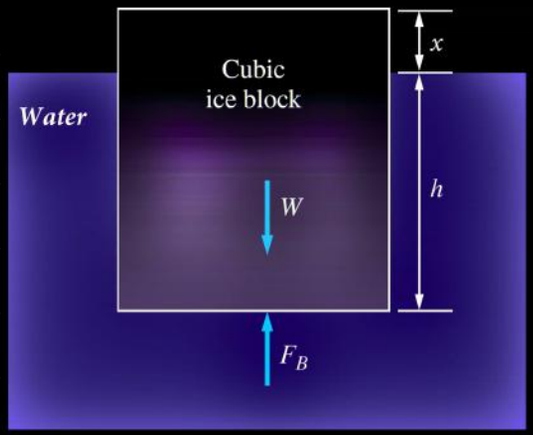


S: Specific gravity of floating body
S_ℓ: " " " liquid.

For ice, $S = 0.9$ thus 90% volume of ice remains below the free surface of water.

Effect Of Melting Of Floating Ice On Free Surface Of Water:

Ice is nothing but water in frozen form. The weight of water displaced (i.e water filled in imaginary hollow created by the ice in the water body) is equal to the weight of ice. Thus upon melting, the ice converts to water just sufficient to fill the imaginary hollow (before melting of ice). Thus melting of ice has no effect on free surface of water. The surface level remains exactly same.



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→ This is true for ice floating in water (not in arbitrary liquids)
→ This is assuming floating ice. If ice were experiencing normal reaction from vessel floor, things would be different.

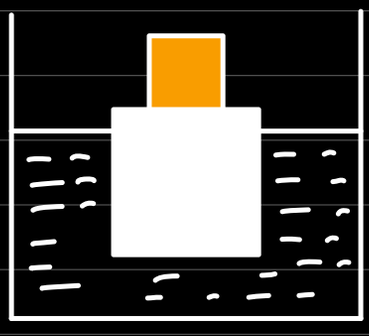
Think! What would have happened if instead of freely floating, ice experienced some normal reaction from container's floor? (Not watertight).

Solution: In that case, weight of ice is supported by weight of water displaced + the normal reaction. Thus weight of water in the imaginary hollow created by ice is less than the weight of ice itself. Thus, the amount of water produced by melting the ice can't be accommodated in that hollow and therefore, surface level will rise upon melting of ice.

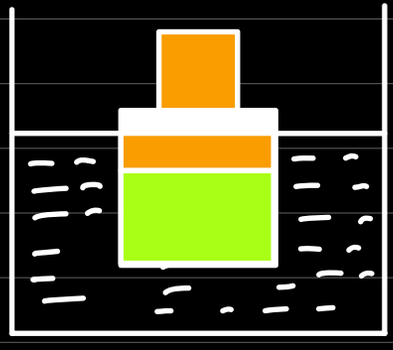
Think! What would happen if ice were floating in a liquid denser than water?

Solution: In that case the volume of "hollow" created by ice will be less than that created in water (Because smaller volume of this liquid will weigh equal to ice as compared to that of water). Thus melted ice can't be accommodated in that hollow and therefore, the level of free surface will increase upon melting of ice.

EXAMPLE: A block of ice supports a wooden block while floating in water. What happens to the water level after all the ice melts.

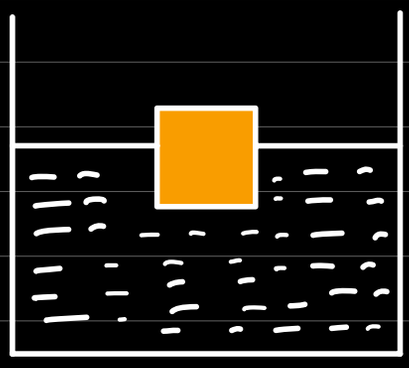


Solution: The total displaced volume i.e the "hollow" in water has a weight equal to combined weight of ice and wood. We can mentally divide this hollow in to two parts corresponding to the volumes required for supporting wood and ice separately. (Say green parts represents volume displaced for ice and orange the wood).



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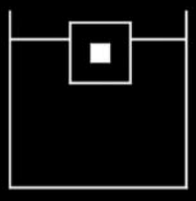
Now when the entire ice melts, the molten ice exactly fills the green volume and wood block still floats displacing a volume corresponding to its weight (which is same as before). Thus there is no net change in volume under the free surface accordingly level of free surface still remains the same.



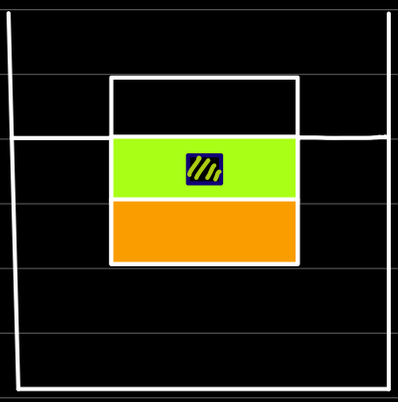
EXAMPLE:

There is a metal cube inside a block of ice which is floating on the surface of water. The ice melts completely and metal falls in the water. Water level in the container

(A) Rises
(B) Falls
(C) Remains same
(D) Nothing can be concluded



Solution: The weight of total displaced volume must be equal to the sum of weights of the metal and ice. Once again let us mentally divide the hollow in to two displaced volumes corresponding to balancing the weight of metal and ice (say respectively green and orange volumes). Since metal is denser than water. The volume displaced to balance the metal is more than the volume of the metal itself. Now when the entire ice melts, entire ice fills the orange volume and now metal sinks to the bottom thus displacing the liquid equal to its actual volume (which is less than the green volume). Thus the final displaced volume under the free surface is less than initial displaced volume. Therefore the level of free surface of liquid must go down.



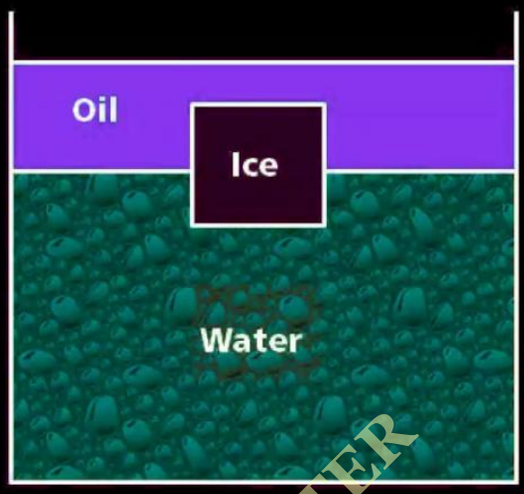
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Example:

Challenge : An Ice Cube is Floating in Water over which a layer of oil is poured as Shown in the Figure. Now if all the ice melts:

(a) What will happen to the lower level of oil water interface? Will it rise fall or remain at the same level.

(b) What will happen to the upper surface of the oil? Will it rise , fall or remain at the same level?



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Solution: Weight of ice is equal to sum of weights of displaced oil and water. Thus weight of water displaced must be less than the weight of ice and accordingly when ice melts, the entire molten volume can't be accomodated in the "hollow" inside water volume. Thus some of molten fluid will form a layer above water surface. Hence the lower oil water inteface will rise. Now since total volume in the container has decreased upon melting of ice. Therefore upper level must go down.

Work Done In Moving A Submerged Object:

Work required to move a submerged object from an initial position to a final position can be conveniently calculated by considering the gravitational potential energy changes of liquid and the submersed object in the process of moving the object. For visualization purposes we typically follow these steps:

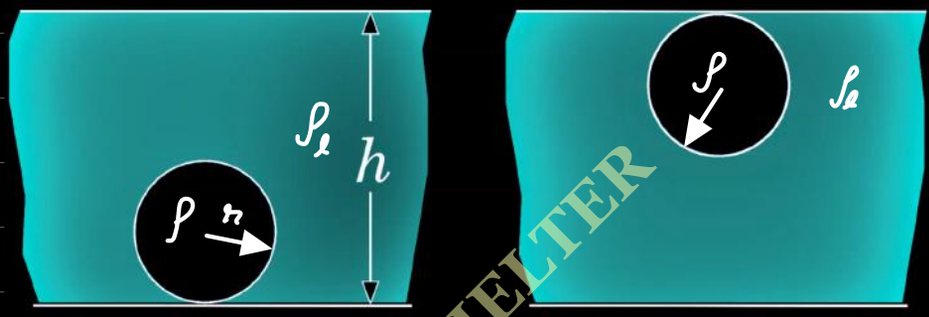
1. Make the object disappear making us visible a hollow
2. Make the liquid from final position of object disappear thus making another hollow.
3. Make the disappeared liquid appear in the hollow created by the object and if the hollow is not completely filled by the disappeared liquid, scrape additional liquid from surface to

fill the remainings hollow.

4. Make the object appear in the hollow created by disappeared liquid.

We can classify the problems involving moving submerged object in to two categories:

Object Remains Fully Submerged Upon Moving:



Consider raising of a spherical body from floor to the surface as shown in the figure. Here as the sphere moves to the top, from potential energy point of view it is as if the position of the solid sphere is swapped with a liquid sphere (which occupies its previous location). Thus work done in slowly changing the position of the sphere:

$$\Delta W = \Delta U = mg\Delta y - m_w g \Delta y$$

m: mass of sphere, m_w: mass of water displaced, Δy: change in y coordinate of CM of sphere (assuming uniform density).

Thus:

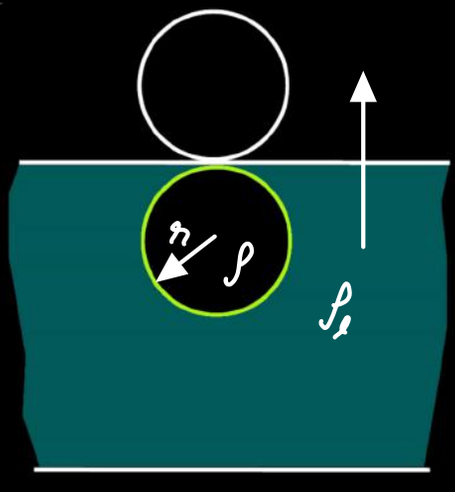
$$\Delta W = \left(\rho \times \frac{4}{3}\pi r^3 - \rho_w \times \frac{4}{3}\pi r^3\right)[h - 2r] = (\rho - \rho_w) V [h - 2r].$$

This could also be seen directly as work done against resultant of weight and buoyant force.

Bringing An Object Just Below The Surface To Just Above The Surface:

Consider a subsurface sphere being brought above the surface as shown in the figure. Here when the sphere is brought up, its hollow is imagined to be filled by scraping molecules from surface of the pool. Thus, work done in process:

$$\Delta W = \Delta U = \rho \times \frac{4}{3}\pi r^3 g \times 2r - \rho_l \times \frac{4}{3}\pi r^3 g \times r$$



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→ For a general movement of the bodies we can either break the movements in to superposition of the above two cases or directly visualize the change in potential energy changes by "filling the hollow" visualization.

EXAMPLE: PATHFINDER MCQ-2:

A sphere of mass m and radius r rests at the bottom of a large reservoir of water as shown in figure-I. Depth of the reservoir is h. Density of material of the sphere is the same as that of water. Now the sphere is slowly pulled completely out of water as shown in figure-II.

Which of the following is a correct expression for work done by the agent pulling the sphere?

(a) mgr (b) $0.5mgr$
(c) $mg(r+h)$ (d) $mg(0.5r+h)$

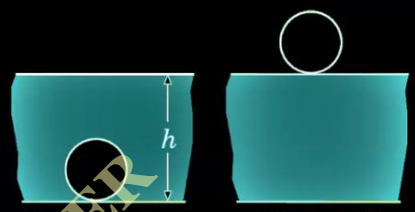
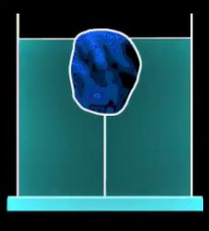


Figure-I Figure-II

Solution: Since densities of liquid and sphere are identical there is no work done until the sphere is just below the surface. After that gain in PE of sphere : $\Delta U_s = mg \times 2r$, loss in PE of water (scraping surface molecules to fill the hollow), $-\Delta U_l = mgr$, thus net work done : $\Delta W = \Delta U_s + \Delta U_l = mgr$ i.e option (a) Ans.

EXAMPLE: PATHFINDER: MCQ 7:



A piece of ice tied with the help of a thread to bottom of a cylindrical container having some amount of water. Length of the thread is so adjusted that the ice is not fully submerged. Tension force in the thread is 1.0 N. How much and in what way will the water level in the container shift, when all the ice melts? Area of the bottom of the vessel is 400 cm^2, density of water is 1.0 g/cm^3 and density of ice is 0.9 g/cm^3.

(a) 2.5×10^{-3} m upwards (b) 2.5×10^{-3} m downwards
(c) 2.8×10^{-3} m upwards (d) 2.8×10^{-3} m downwards

Solution: Here also we can mentally divide the submerged portion of ice in to two parts (a) To balance its own weight (b) Additional submergence to balance the tension. The entire ice will be accomodated in displaced volume corresponding to its weight and the displaced volume corresponding to additional submergence will go to zero upon melting. Now additional submerged volume (V_a) is given

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by the equation: $T = \rho_w V_a g \Rightarrow V_a = T/\rho_w g = 1N/1000\times 10$

$\Rightarrow V_a = 10^{-4}\ m^3$, Thus decrease in water level, $\Delta y = V_a/A$ where A is base area i.e $\Delta y = [10^{-4}/(400\times10^{-4})]m = 2.5\times10^{-3}\ m$ downwards.

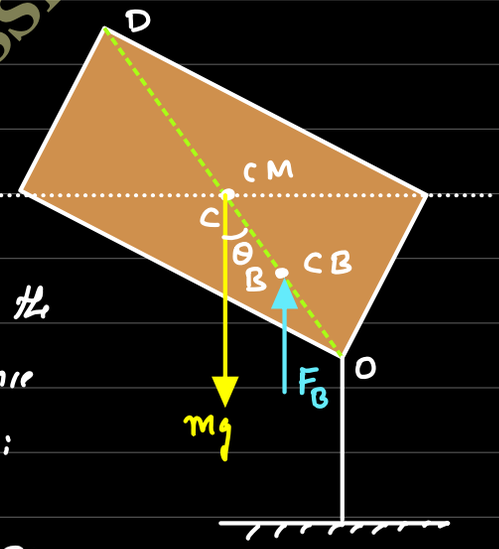
Example: Pathfinder MCQ 9:



A wooden plank is floating on water in a pool. It is tethered to the bottom of the pool by a string attached at mid-point of an edge of its bottom face, which causes it to float with a diagonal of one of its vertical cross-section coinciding with the level surface of the water, as shown in the figure. What is the specific gravity of the wood?

(a) 1/5 (b) 1/4
(c) 1/3 (d) 1/2

Solution: For ease of visualisation, we have redrawn the plank with a more convenient aspect ratio. The weight acts through CM of plank while buoyant force acts through centre of buoyancy of submersed part (i.e centroid of the lower triangle). Now if $OC = \ell$, $OB = 2\ell/3$ since B is centroid. Now balancing torques about O:



$\rho_p V g\, \ell \sin\theta = \rho_w \frac{V}{2} g \left(\frac{2\ell}{3}\right) \sin\theta$ (ρ_p: Plank density).

Thus $\frac{\rho_p}{\rho_w} = \frac{1}{3}$ Ans. i.e option (C).

Example: Pathfinder: Build 3:

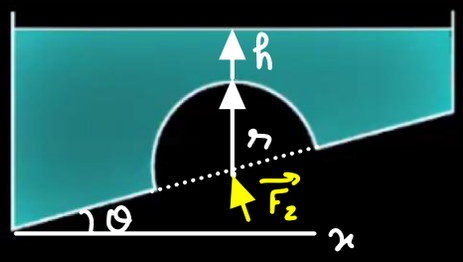


Bottom of a vessel is inclined at an angle θ and has a hemispherical dent of radius r as shown in the figure. If a liquid of density ρ is filled in the vessel to a height h above the highest point of the dent, find a suitable expression for magnitude of force of liquid pressure on the dent excluding contribution of the atmospheric pressure.

Solution: Here we will use compensation idea for water tight surface in the buoyant force. Had there been no water tight surfaces, the buoyant force experienced by the hemisphere:

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$\vec{F_1} = \rho \times \frac{2}{3}\pi r^3 g\,\hat{j}$ Now the missing force



on water tight surface :

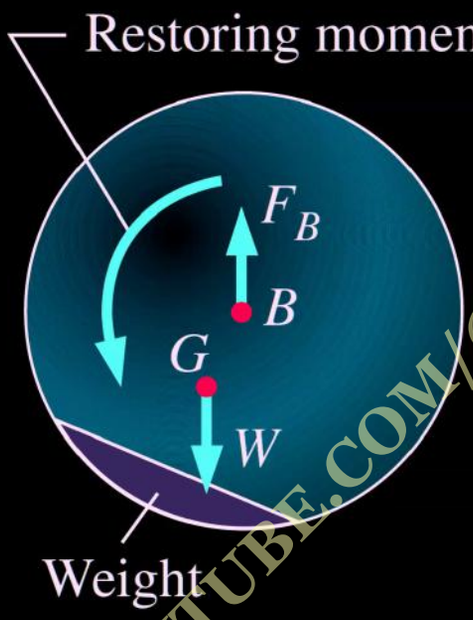
$\vec{F_2} = \rho g (h+r)\,\pi r^2 [\cos\theta\,\hat{j} - \sin\theta\,\hat{i}]$

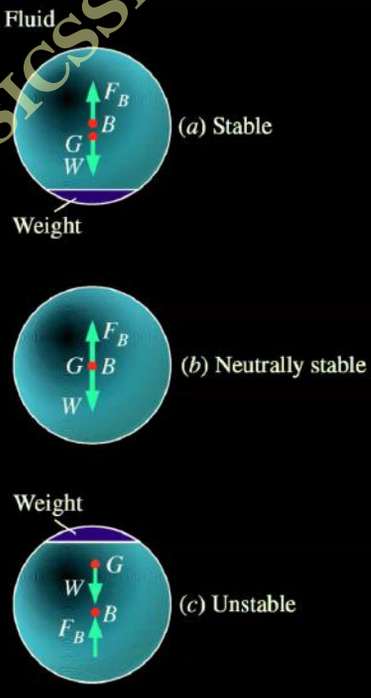
Thus net force on hemispherical surface :

$$\vec{F} = \vec{F_1} - \vec{F_2} = \rho\pi r^2 g \left[(h+r)\sin\theta\,\hat{i} + \left(\frac{2}{3}r - (h+r)\cos\theta\right)\hat{j}\right]$$ Ans.

$$|\vec{F}| = \left[\left(\frac{2\pi r^3 \rho g}{3}\right)^2 + \left(\pi r^2 \rho g (h+r)\right)^2 - 2\left(\frac{2\pi r^3 \rho g}{3}\right)\left(\pi r^2 \rho g (h+r)\cos\theta\right)\right]^{1/2}$$ Ans.

STABILITY OF NEUTRALLY BUOYANT BODIES

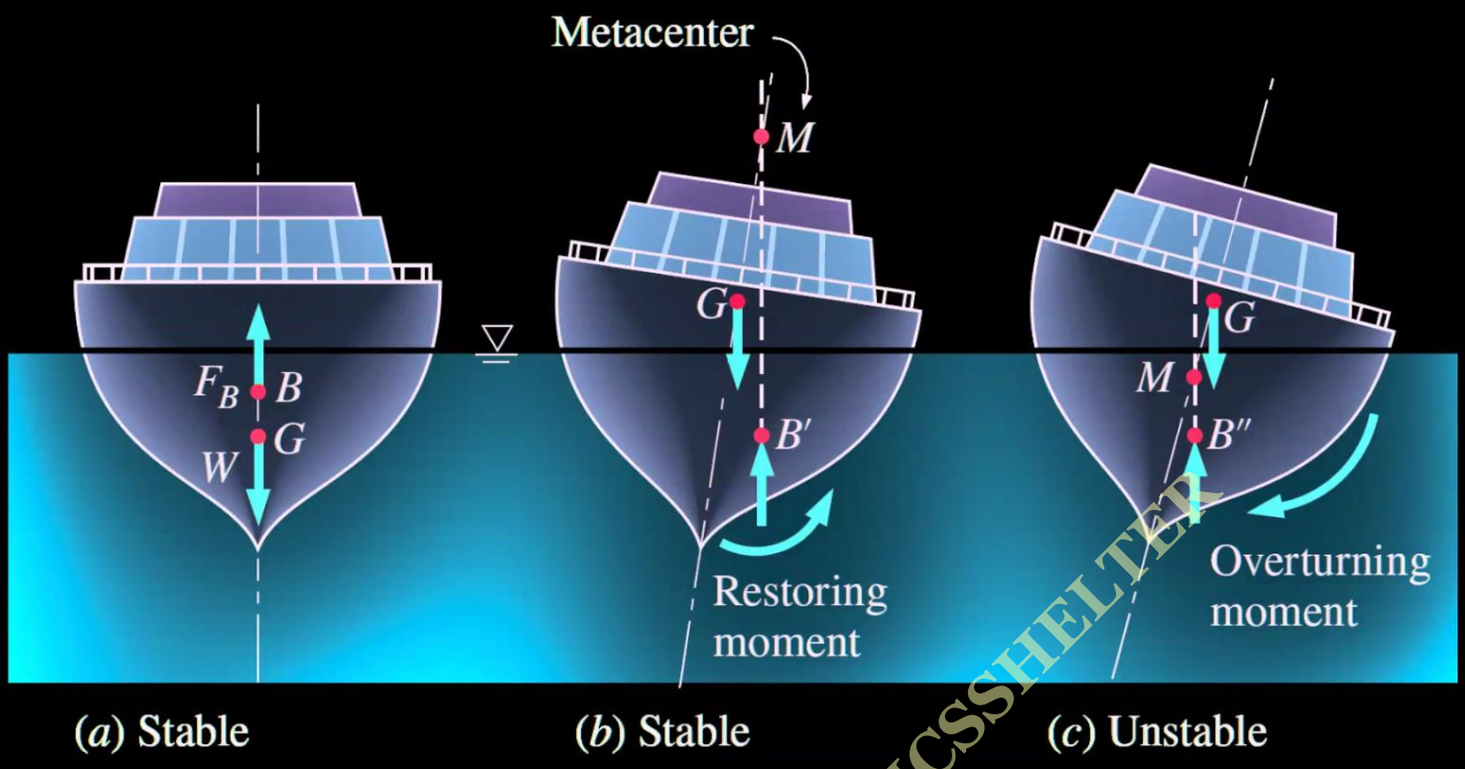




A neutrally buoyant body is one whose average density is equal to that of the fluid in which it is submerged. For stability, the body must experience a restoring torque upon being turned through some angle. (i.e. if the body rotates clockwise about its CM, it must experience an anti-clockwise torque). As can be inferred from the left figure above, for a neutrally buoyant body, for stability centre of mass must be below the centre of gravity.

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Stability Of Floating Bodies:



Metacentre: For fully submerged rigid bodies, the position of centre of buoyancy does not change relative to the rigid body as rigid body is turned inside water, however for floating bodies in general, the shape of displaced liquid itself changes upon turning the floating body. Suppose a ship floats in equilibrium, then the line BG must be vertical for zero torque (B: centre of buoyancy, G: centre of gravity). Now suppose the ship is given a small roll keeping the fraction of submerged volume constant, then the centre of buoyancy shifts relative to ship say to a point B'. The intersection point of line BG (rigidly attached to ship) with vertical through B' is called metacentre. For stability, Metacentre must be above G while if metacentre is below G, the ship is in unstable equilibrium.

(This simply means that if ship tilts rightwards, for stability, Buoyant force line must also shift rightwards relative to G to provide restoring torque about G). Higher the metacentric height, greater the stability.

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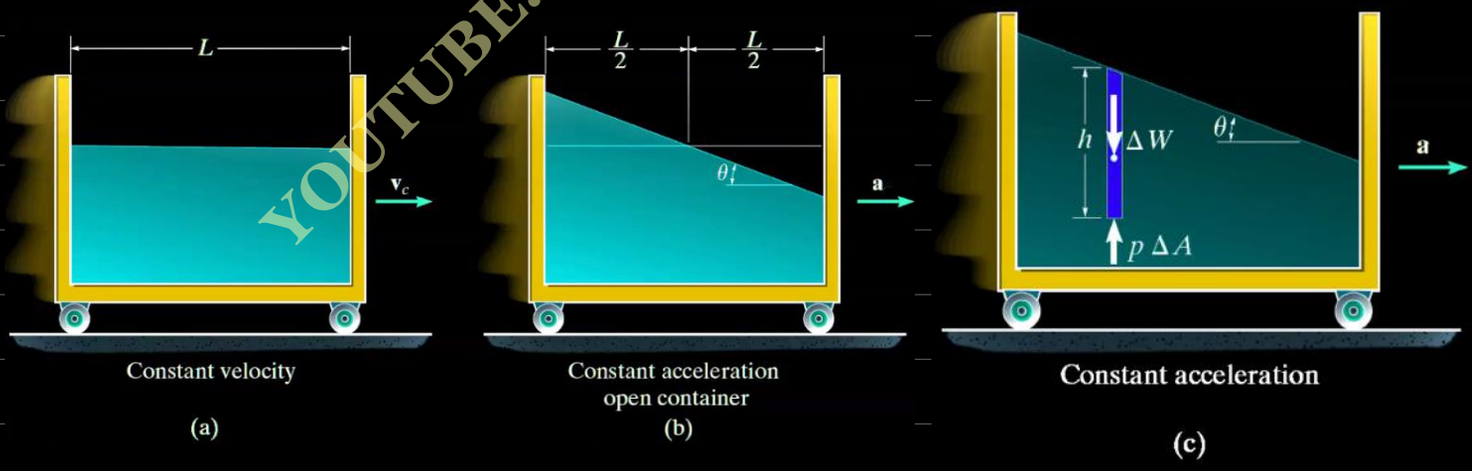
Stability Of Bodies With Negligible Waterline:



Suppose a floating body has negligible waterline around it. In that case as the body turns both centre of Buoyancy and centre of gravity turn with it. Imagine turning such a body about point O on the waterline, then for stability, torque of gravity must be more than that of Buoyant force. Thus for stability, in this case also, centre of Buoyancy must be above the centre of gravity.

Constant Translational Acceleration Of A Liquid:

Constant Horizontal Acceleration:



Consider vertical column of fluid element in a horizontally accelerated fluid as shown in the figure. Here the weight of the column must be supported by the gauge pressure force. Accordingly, $P\Delta A = \Delta W$

$\Rightarrow P\Delta A = \rho \Delta A h g \Rightarrow P = \rho g h$ ①

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Now consider a horizontal element of fluid. The pressure forces on this element must provide the horizontal acceleration.

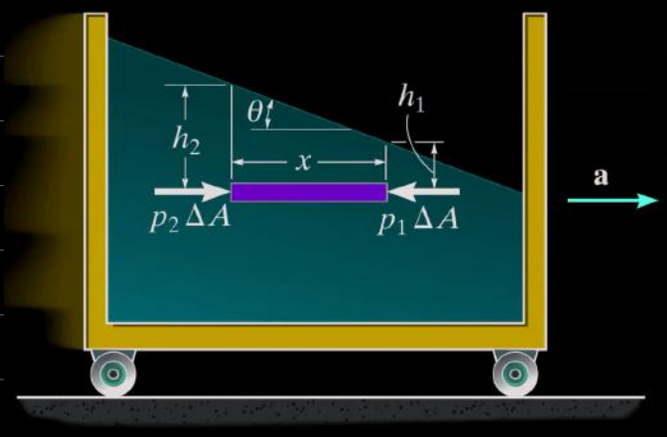
Thus $(P_2 - P_1)\Delta A = \rho \Delta A x a$

$\Rightarrow \quad P_2 - P_1 = \rho a x$

Now $P_2 = \rho g h_2$ and $P_1 = \rho g h_1$

Thus $\rho g (h_2 - h_1) = \rho a x \Rightarrow \frac{h_2 - h_1}{x} = \frac{a}{g}$

$\Rightarrow \quad \boxed{\tan\theta = a/g}$



Constant acceleration

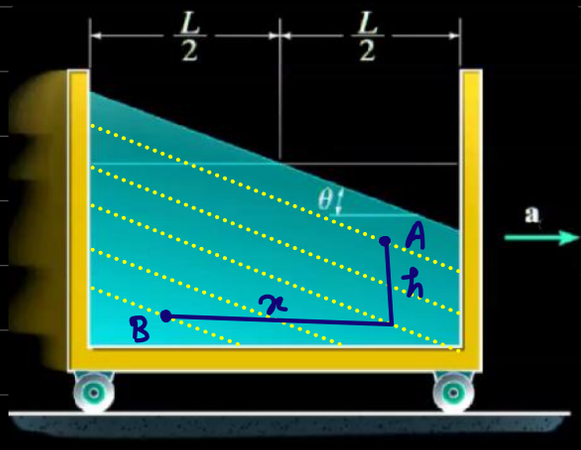
(d)

→ Here acceleration of the vessel can be modeled as pseudo gravity in horizontal direction. Just as pressure changes in vertical direction as $\rho g h$ (increases along gravity) similarly in horizontal pressure increases along pseudo gravity (leftwards in our case).

Isobars:

The lines of constant pressure are known as isobars. In case of translationally accelerated vessels, isobars are parallel to the free surface of the liquid (as shown by dotted yellow lines in the adjacent figure).

(At points of equal depth from free surface, pressure is same).

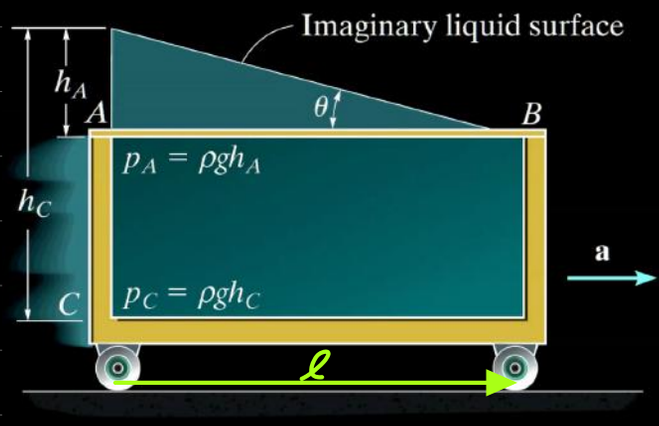


→ Based on our discussion from this section, we can relate pressure at points A and B as $P_B = P_A + \rho g h + \rho a x$.

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Pressure Variation In A Completely Filled Closed Container:

Suppose a container is fully filled with a liquid and closed with a lid (The least gauge pressure being zero (Imagine a very tiny bubble at atmospheric pressure at the top of container)). Now if the container accelerates as shown, the least pressure will occur at B (∵ pressure increases leftwards as well as downwards). Thus gauge pressure at A will be $\rho a l$. Had there been no lid, the height difference between A and B would be $h = l \tan\theta$ where $\tan\theta = a/g$. Thus $\rho a l = \rho a l \tan\theta \frac{g}{a}$

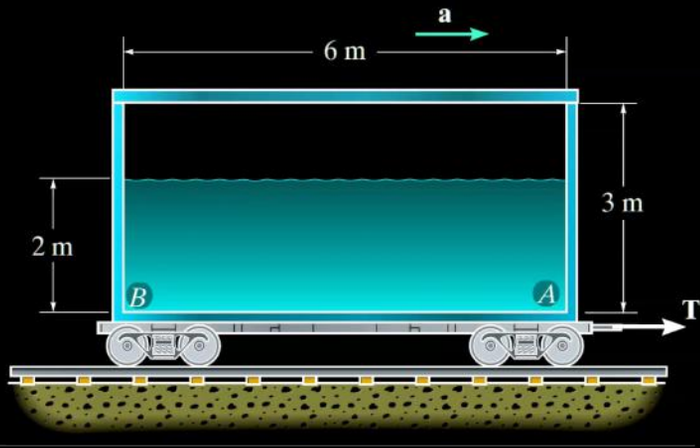


Constant acceleration of a completely filled closed container

i.e $P_A = \rho g h$. Thus pressure at A is as if there were an imaginary liquid surface as shown in the figure.

Example:

The closed railroad car is 6 m wide and filled with water to the level as shown. Find the gauge pressure at A and B when car moves with a constant acceleration $a = 4\ m/s^2$. Take $g = 10\ m/s^2$.

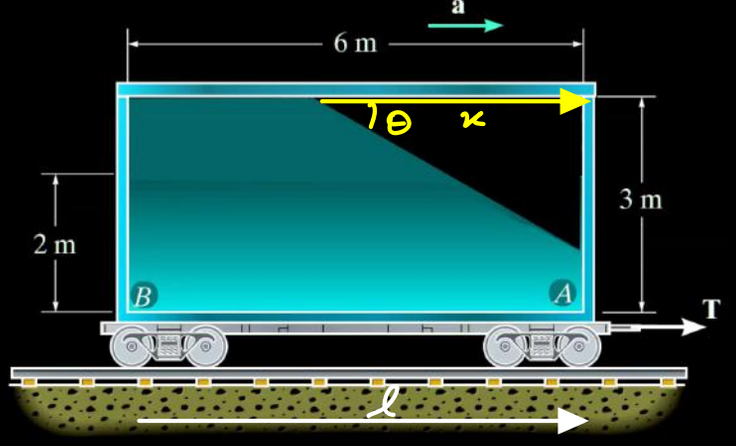


Solution: $\tan\theta = a/g = 2/5$

Consider the adjacent figure. By volume conservation, $\frac{1}{2} x \cdot x \tan\theta = 1 \times 6$

$\Rightarrow \quad x^2 = 30 \Rightarrow x = \sqrt{30}$

Now $P_A = \rho g (3 - \sqrt{30} \tan\theta)$



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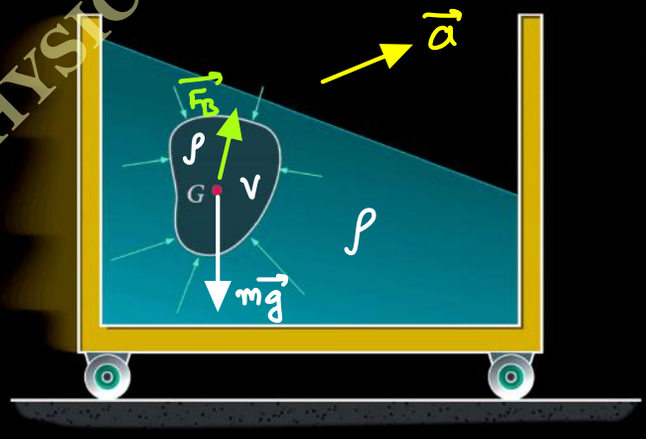
$= 1000 \times 10 (3 - 2.19) = 8.1 \times 10^3$ Pascals.

Now $P_B = P_A + \rho a l = 8.1 \times 10^3 + 10^3 \times 4 \times 6 = 3.21 \times 10^4$ Pascals Ans.

→ If the acceleration were large enough, the empty volume would transform in to a trapezium shape from triangular and pressure at A will become zero (Think why).

→ In case of acceleration in a random direction we can consider its components in vertical and horizontal direction and find the angle of surface using concept of effective gravity. For our figure in that case, $\tan\theta = a_x/(a_y+g)$ [x: left y: upwards]

Buoyant Force In Accelerated Vessels:



Consider a body submersed in a vessel having acceleration in arbitrary direction. The buoyant force experienced by this must be same as that experienced by a water volume of identical shape. This is because colliding molecules don't know what material they are going to collide with. Now consider the FBD of such volume of water. From ground frame, the resultant of its weight and buoyant force must be sufficient to accelerate this fluid body thus:

$$\rho V\vec{g} + \vec{F_B} = \rho V\vec{a}, \text{ thus}$$

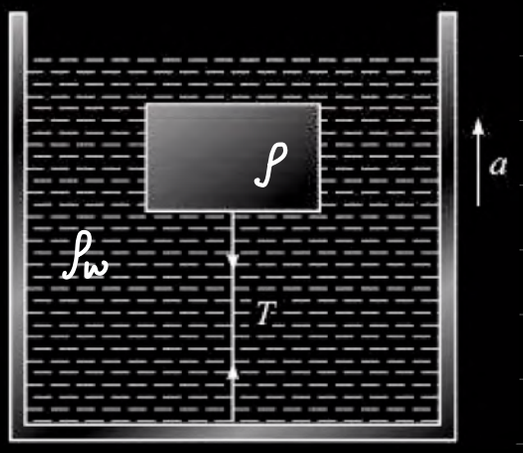
$$\boxed{\vec{F_B} = \rho V\vec{a} - \rho V\vec{g}}$$ ← Net Buoyant force in accelerated Vessels.

→ Notice that second term represents the usual buoyant force that occurs in stationary vessels.

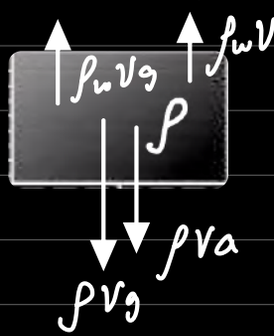
→ The first term on RHS represents additional buoyant force that occurs due to acceleration of the vessel. Notice that it is directed along the acceleration of the vessel (i.e against pseudo gravity).

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EXAMPLE: A block of volume V and density ρ is tied to floor of a vessel by a massless string as shown. If now vessel accelerates upwards, will the string tension increase or decrease? Also find acceleration of the block in vessel frame if the string is cut.



Solution: The additional forces introduced due to acceleration of fluid are the additional buoyant force and the pseudo force and the change in tension occurs to balance these additional forces. (Clearly $\rho < \rho_w$ from the figure). Now additional Buoyant force, $\vec{F}_{BA} = \rho_w V a \uparrow$ and the pseudo force is $\rho V a \downarrow$. Now since $\rho < \rho_w$, additional buoyant force exceeds the pseudo force in downward direction. Thus tension which acts downwards must increase to retain the balance. Infact change in tension $\Delta T = (\rho_w - \rho) V a$. Now when the string is cut, we can find the acceleration in vessel frame considering the FBD. Thus we have four forces namely weight, pseudo force, Regular Buoyant force and additional buoyant force due to acceleration of the vessel.



Thus, if a_0 is acceleration of block in vessel frame:

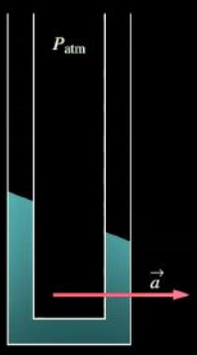
$$\rho_w (a+g) V - \rho (a+g) V = \rho V a_0 \text{ thus,}$$

$$a_0 = (\rho_w - \rho)(a+g)/\rho \uparrow \text{ Ans.}$$

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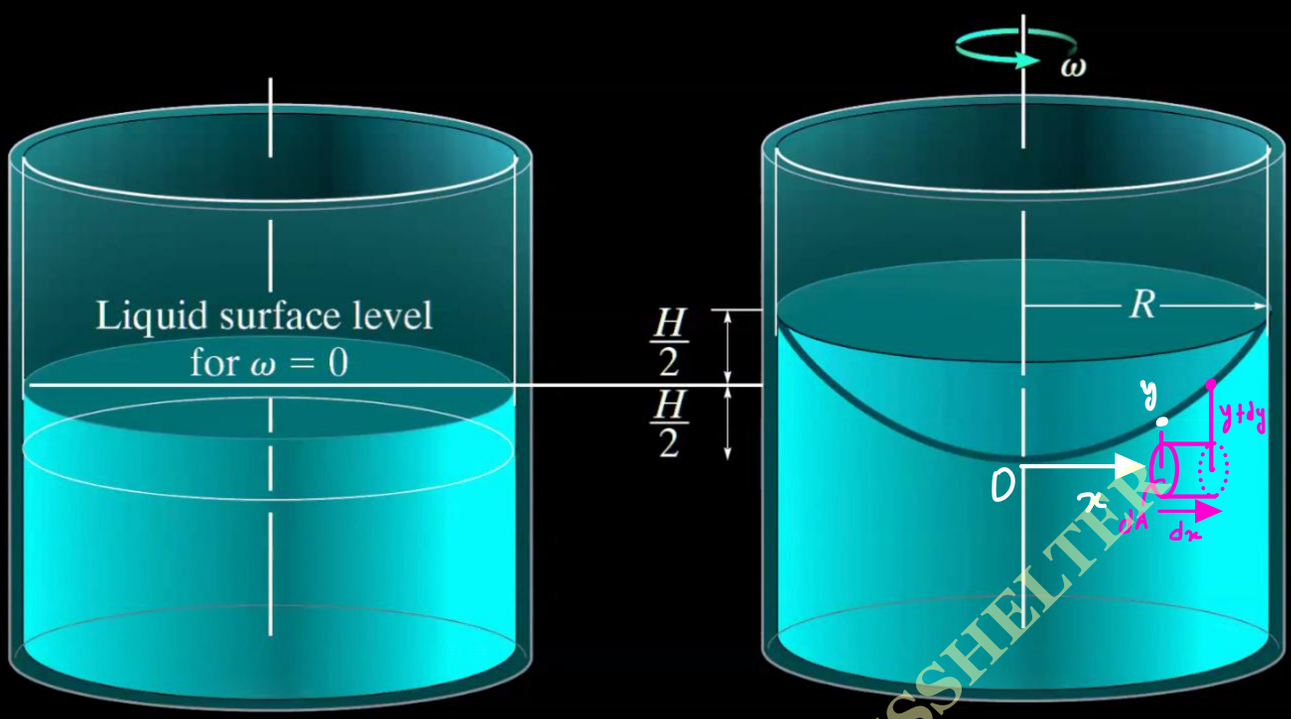
INCLINATION OF FREE SURFACE IN ARBITRARY VESSEL:

Using the concept of effective gravity it can be readily seen that in any arbitrary vessel even with discontinuous fluid surface (such as U tube) the inclination of surface is given as $\tan\theta = a/g$. (The free surface patches are all coplanar).



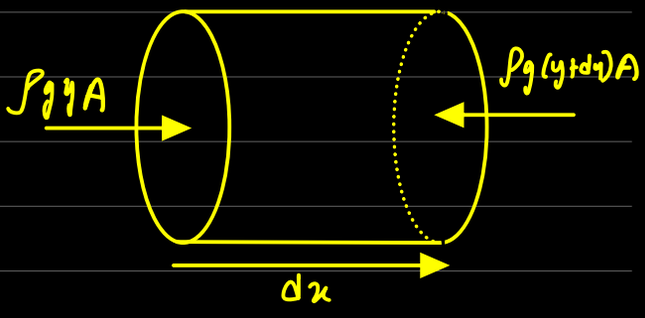
When a continuous liquid in a container of any shape (even a U-tube manometer!) is accelerated at constant rate, the liquid behaves as a rigid body with a tilted plane surface. The surface itself does not have to be continuous.

Steady Rotation Of A Liquid:



When a fluid in a container is rotated about a fixed axis, the free surface of the fluid assumes a paraboloid shape such that highest and lowest points of paraboloid are equidistant from free surface when liquid was not rotating. (This can be seen using the fact that volume of a paraboloid is exactly half of its enveloping cylinder). We will prove the parabolic shape considering FBD of a fluid element. As shown in the figure we consider a small cylindrical fluid element of cross section dA extending from x to $x+dx$ with depths of its circular faces being y and $y+dy$ from free surface.

The resultant of pressure forces must be sufficient to provide the centripetal acceleration to the fluid element.



Thus,

$$\rho g (y+dy) A - \rho g y A = (\rho A dx)\omega^2 x$$

$$\Rightarrow \quad g\,dy = \omega^2 x\,dx \quad \Rightarrow \quad \int_0^y g\,dy = \int_0^x \omega^2 x\,dx$$

$$\Rightarrow \quad \boxed{y = \frac{\omega^2 x^2}{2g}} \leftarrow \text{Equation of Rotating fluid's free surface}$$

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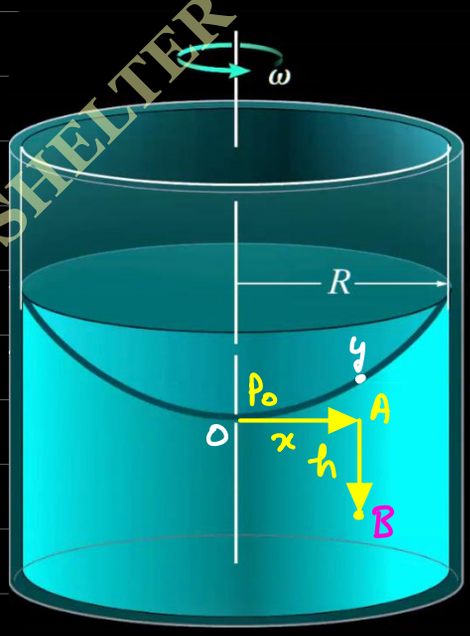
This is clearly an equation of parabola with vertex on rotation axis.

→ In case of rotating vessels of arbitrary shapes also the shape of the free surface is part of a paraboloid with vertex on axis of rotation (vertical). The axis may either be in the vessel or could also pass from outside the vessel.

PRESSURE AT ARBITRARY POINT IN ROTATING VESSEL:

Based on our analysis from previous section, we can write that, $P_A = P_0 + \rho g y$

i.e $P_A = P_0 + \rho \frac{\omega^2 x^2}{2}$ Now accordingly, pressure at B is $P_B = P_A + \rho g h = P_0 + \rho \frac{\omega^2 x^2}{2} + \rho g h$



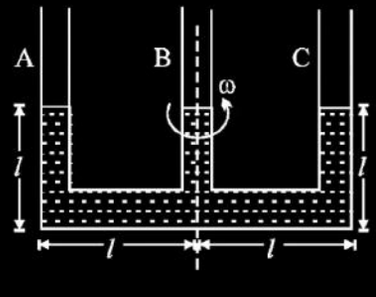
→ If the vessel is rotated sufficiently fast, the vertex of parabola will touch the floor of the vessel and if we rotate it still faster, we will begin to see a circular patch without any fluid. (Now the upper level of fluid must be calculated carefully considering fluid volume conservation).

→ In case of vessel with a lid, we can assume an imaginary fluid surface on top of the vessel for pressure calculations just as we did for translationally accelerated vessels. The difference here is that we will have to imagine a paraboloid surface.

EXAMPLE:

Figure shows a three arm tube in which a liquid is filled upto levels of height l. It is now rotated at an angular frequency ω about an axis passing through arm B. The angular frequency ω at which level of liquid in arm B becomes zero is:

(A) $\sqrt{\frac{2g}{3l}}$ (B) $\sqrt{\frac{g}{l}}$ (C) $\sqrt{\frac{3g}{l}}$ (D) $\sqrt{\frac{3g}{2l}}$



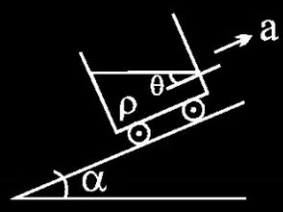
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Solution: The liquid surface will be a part of a paraboloid of the form $y = \omega^2 x^2/2g$. Also when a column length l goes down in middle column, for volume conservation there should be a rise of $l/2$ each in left and right columns. Thus our parabola must pass through $(l, 3l/2)$. Accordingly, $\frac{3l}{2} = \omega^2 \times \frac{l^2}{2g} \Rightarrow \omega = \sqrt{\frac{3g}{l}}$ Ans.

i.e option (C).

EXAMPLE:

A fluid container is containing a liquid of density ρ is accelerating upward with acceleration a along the inclined plane of inclination α as shown. Then the angle of inclination θ of free surface is :



(A) $\tan^{-1}\left[\frac{a}{g\cos\alpha}\right]$

(B) $\tan^{-1}\left[\frac{a+g\sin\alpha}{g\cos\alpha}\right]$

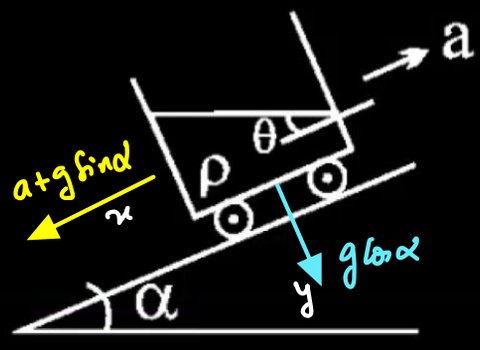
(C) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1+\cos\alpha)}\right]$

(D) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1-\cos\alpha)}\right]$

Solution: Components of effective gravity from vessel frame are as shown in adjacent figure.

Now $\tan\theta = g_{eff\,x}/g_{eff\,y} = \frac{a+g\sin\alpha}{g\cos\alpha}$

Thus correct option is (B).



→ If the container were freely sliding, $a = -g\sin\alpha$, then surface of liquid would be parallel to the incline.

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